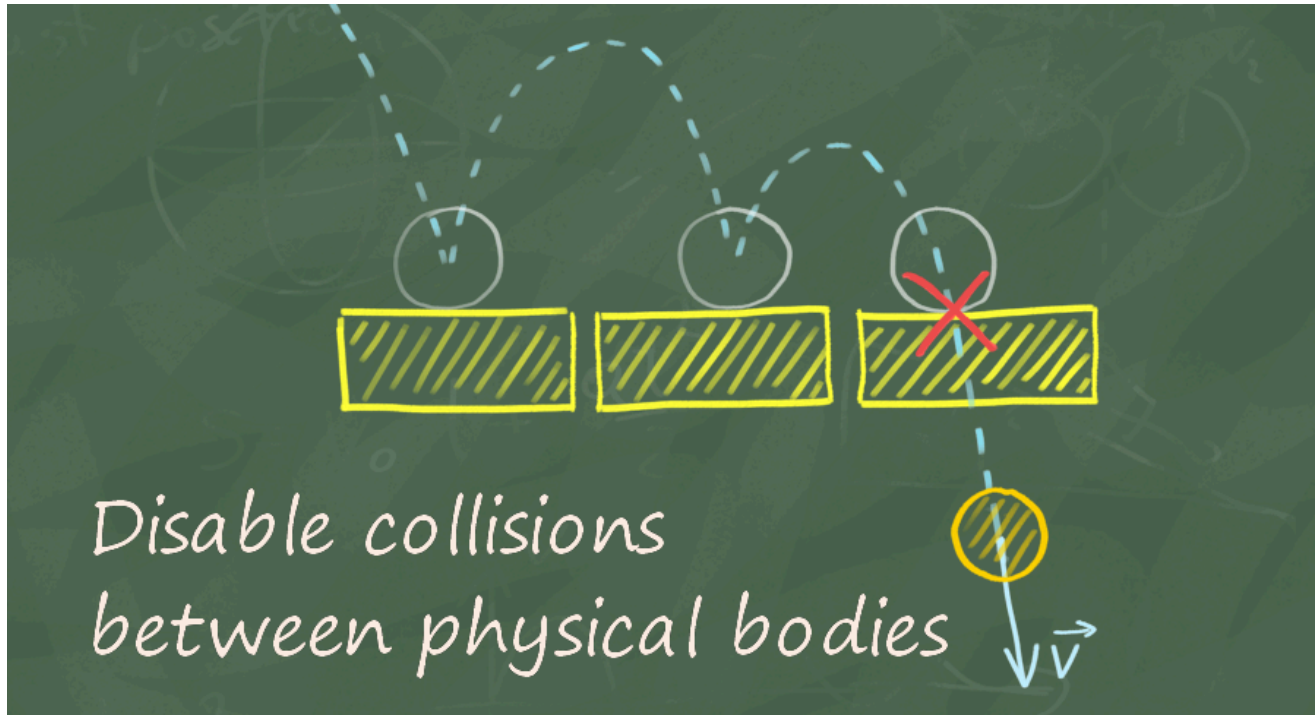


Collision Disabler



Author: **Tomasz Klin**

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Introduction

Thanks to this plugin you will be able to turn off collisions between the indicated pairs of physical objects.

Video: [Collision Disabler](#)

The Unreal Engine provides several methods to disable collisions between objects.

1. [Collision channels](#) switch collisions off between groups of objects, **but not between specific objects**.
2. [Ignore Actor When Moving](#) tells a component to ignore collision with all components of a specific Actor while that component is moved, **but it does not apply to the physics simulation**.
3. [Contact Modification](#) modifies the properties of the generated contact points, but **requires knowledge of PhysX and C++**.

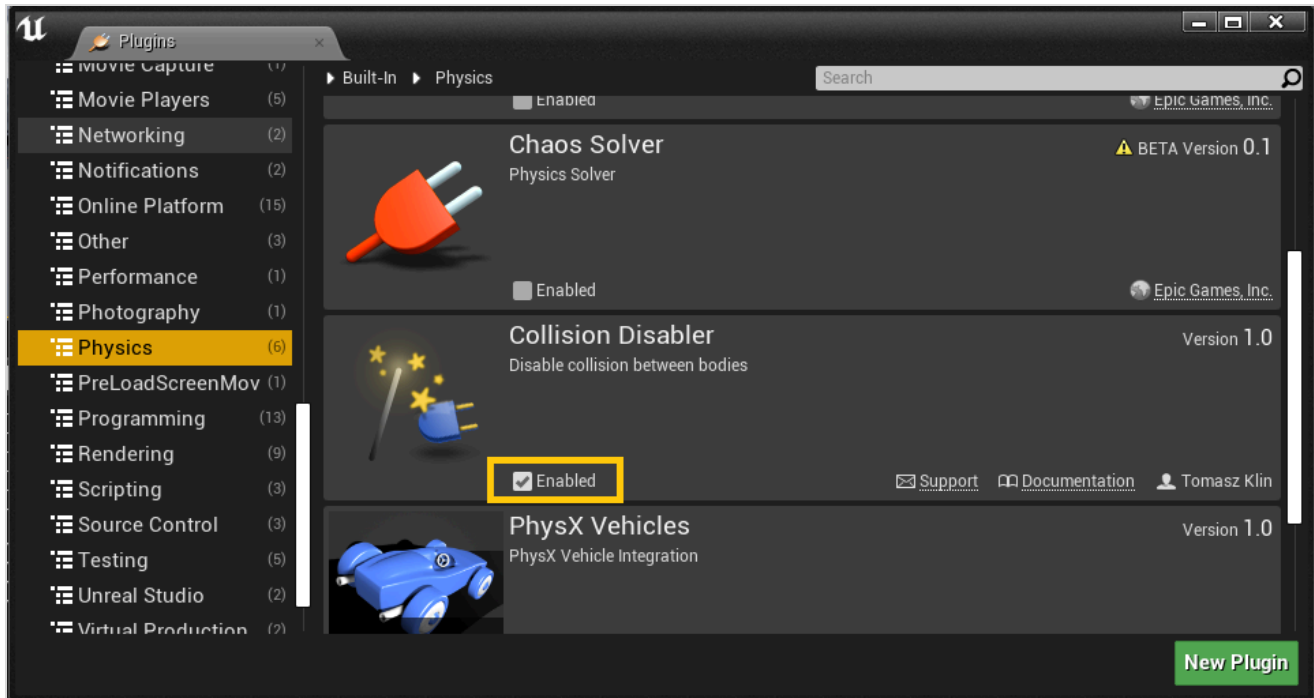
Collision Disabler uses the last method to provide a very simple and convenient way to disable collision between objects using physics simulation.

Features list:

- Disable collision between physical objects
- Works with CCD
- Works with Substepping
- Works with many physical scenes
- Works with both C++ and Blueprint-only projects

Plugin Installation

In order to enable the plugin go to Edit -> Plugins, select **Physics** category and check **Enabled** for Collision Disabler



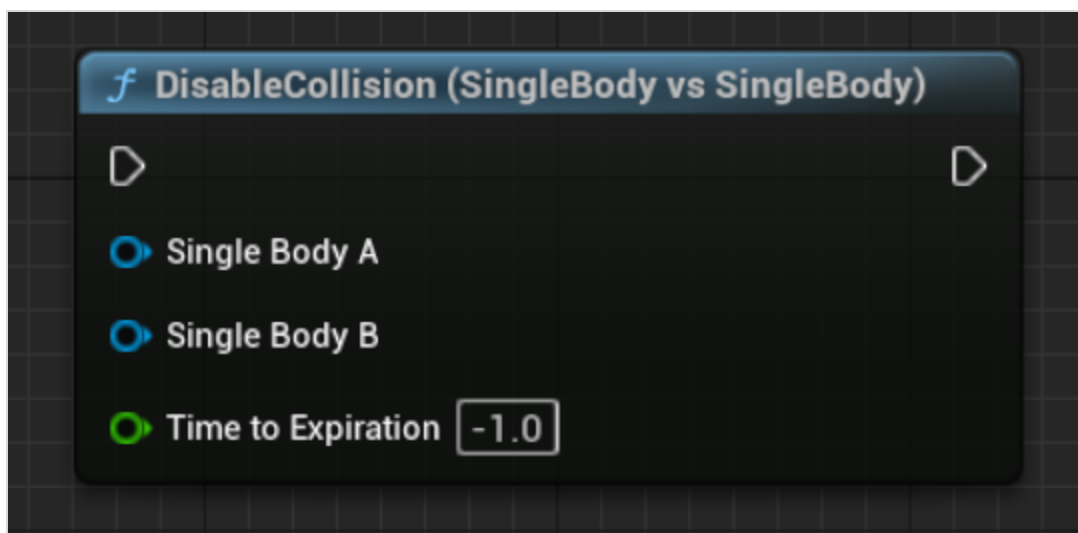
Using in Blueprint

Disable Collision

Single Body vs Single Body

Disable collision between two primitive components with one PhysicalBody e.g. StaticMesh. Pins:

- **SingleBodyA** (UPrimitiveComponent): Input representing the first component for which collisions should be disabled.
- **SingleBodyB** (UPrimitiveComponent): Input representing the second component for which collisions should be disabled.
- **TimeToExpiration**: Input representing an optional duration for disabling collisions (in seconds). After this time, collisions will automatically be re-enabled.

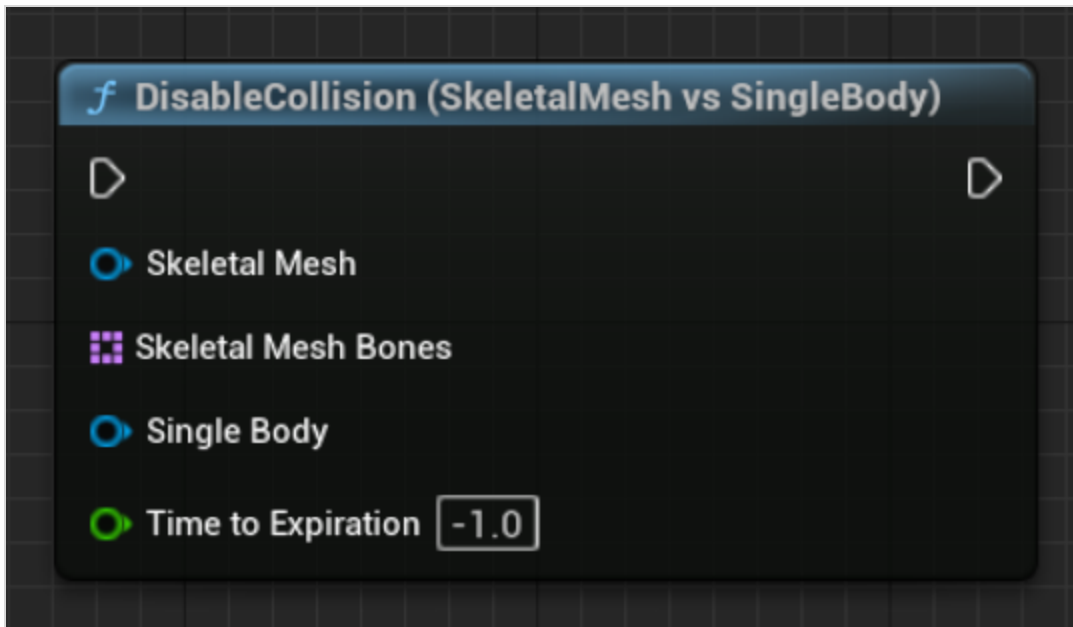


Skeletal Mesh vs Single Body

This node allows you to disable collisions between a SkeletalMeshComponent and a PrimitiveComponent. You have to specify which bones in the SkeletalMeshComponent should be considered when disabling collisions. Pins:

- **SkeletalMesh** (USkeletalMeshComponent): Input representing a SkeletalMesh component for which collisions should be disabled.
- **SkeletalMeshBones** (array of FName): Input representing an array of bone names in the SkeletalMeshComponent

- **SingleBody** (UPrimitiveComponent): Input representing the second component for which collisions should be disabled.
- **TimeToExpiration**: Input representing an optional duration for disabling collisions (in seconds). After this time, collisions will automatically be re-enabled.

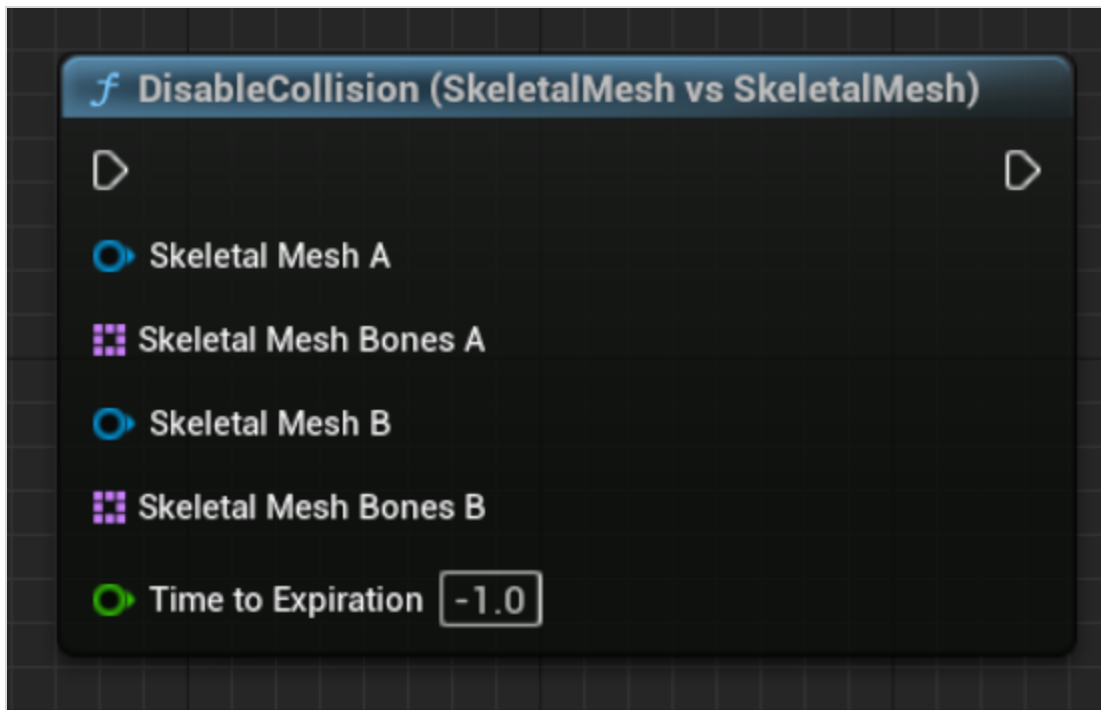


Skeletal Mesh vs Skeletal Mesh

This node allows you to disable collisions between two SkeletalMeshComponents. You have to specify which bones in both components should be considered when disabling collisions.

Pins:

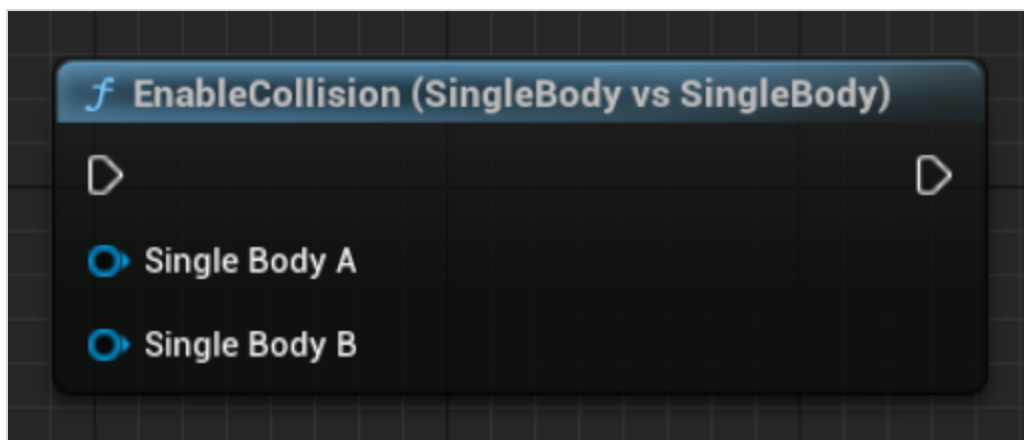
- **SkeletalMeshA** (USkeletalMeshComponent): Input representing the first SkeletalMeshComponent in which collisions should be disabled.
- **SkeletalMeshBonesA** (array of FName): Input representing an array of bone names in the first SkeletalMeshComponent
- **SkeletalMeshB** (USkeletalMeshComponent): Input representing the second SkeletalMeshComponent in which collisions should be disabled.
- **SkeletalMeshBonesB** (array of FName): Input representing an array of bone names in the second SkeletalMeshComponent
- **TimeToExpiration**: Input representing an optional duration for disabling collisions (in seconds). After this time, collisions will automatically be re-enabled.



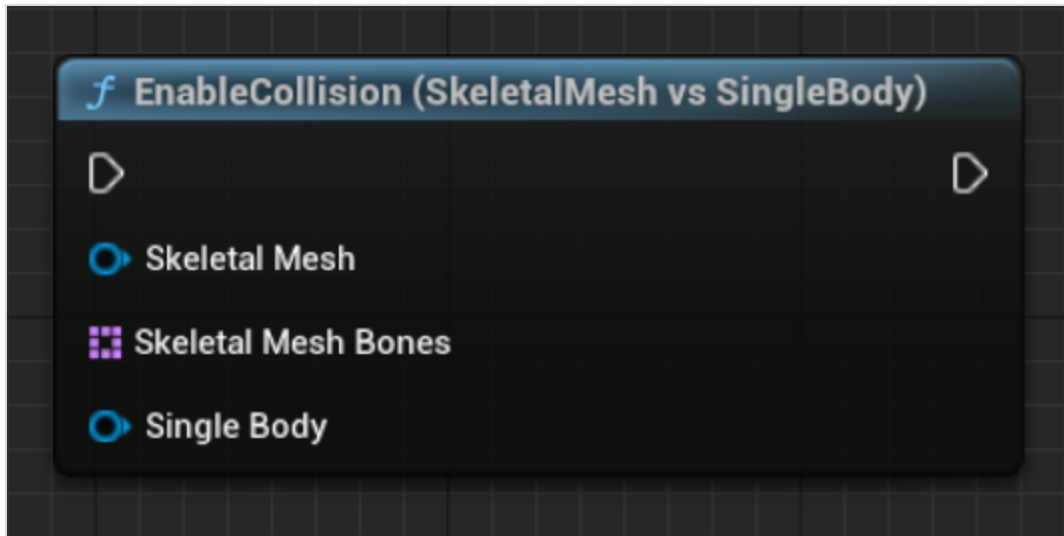
Re-enable Collision

There are corresponding nodes for enabling collisions in the "Collision Disabler" plugin. These nodes allow you to re-enable collisions between the specified components or types of components, essentially reversing the action of the "Disable Collision" nodes. The parameters and functionality of these nodes are similar to the "Disable Collision" nodes, but they enable collisions instead of disabling them.

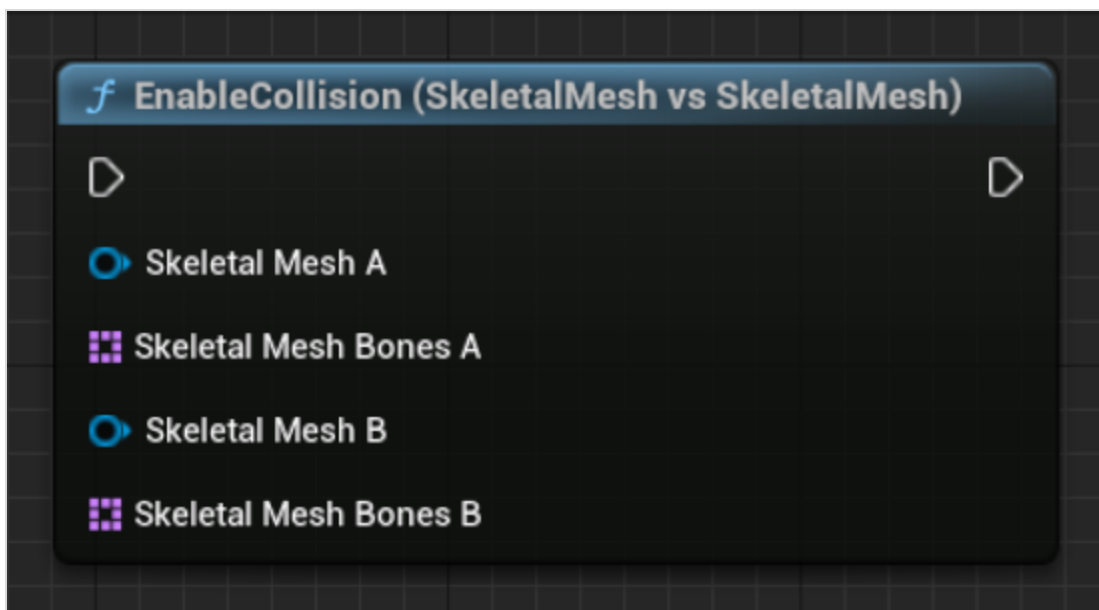
Single Body vs Single Body



Skeletal Mesh vs Single Body

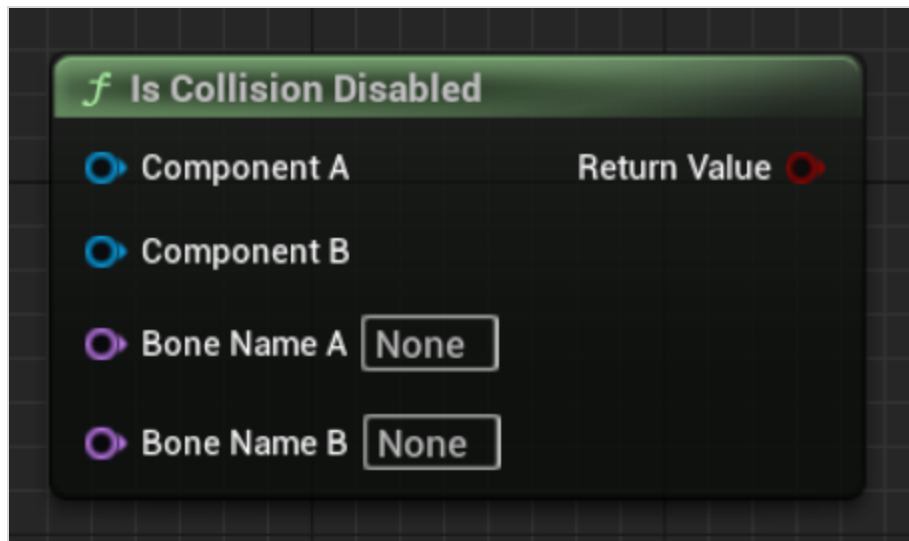


Skeletal Mesh vs Skeletal Mesh



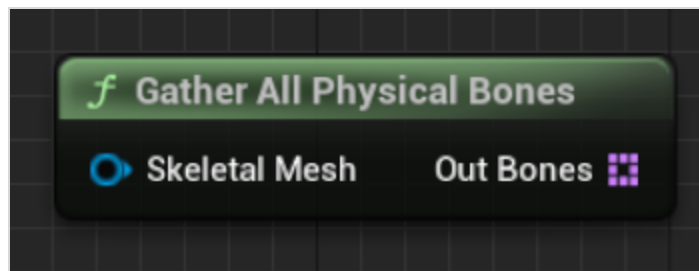
Is Collision Disabled

Check that a given pair of bodies has a disabled collision.



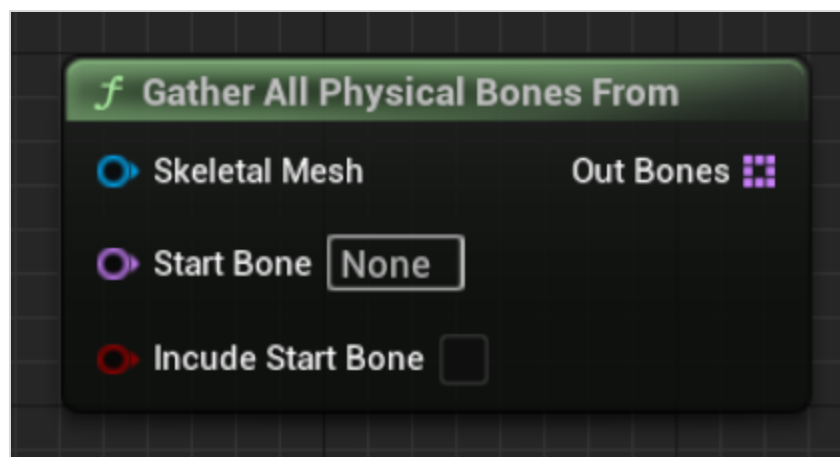
Gather All Physical Bones

This node helps to easily list all bones that have any physics representation



Gather All Physical Bones From

This node returns a list of physical bones from a given bone



Using in C++

Using a plugin in C++ is quite straightforward. All functions described above are located in the **UCollisionDisablerFunctionLibrary** Functions:

Return	Name	Description
void	DisableCollision_SingleBodyVsSingleBody (UPrimitiveComponent* SingleBodyA, UPrimitiveComponent* SingleBodyB, float TimeToExpiration = -1)	Disable collision between two primitive components with one PhysicalBody e.g. StaticMesh.
void	DisableCollision_SkeletalVsSingleBody (USkeletalMeshComponent* SkeletalMesh, const TArray<FName> SkeletalMeshBones, UPrimitiveComponent* SingleBody, float TimeToExpiration = -1)	Disable collisions between a SkeletalMeshComponent and a PrimitiveComponent. You have to specify which bones in the SkeletalMeshComponent should be considered when disabling collisions.
void	DisableCollision_SkeletalVsSkeletal (USkeletalMeshComponent* SkeletalMeshA, const TArray<FName> SkeletalMeshBonesA, USkeletalMeshComponent* SkeletalMeshB, const TArray<FName> SkeletalMeshBonesB, float TimeToExpiration = -1)	Disable collisions between two SkeletalMeshComponents. You have to specify which bones in both components should be considered when disabling collisions.
void	EnableCollision_SingleBodyVsSingleBody (UPrimitiveComponent* SingleBodyA, UPrimitiveComponent* SingleBodyB)	Re-enable collision between two primitive components with one PhysicalBody, such as StaticMesh.
void	EnableCollision_SkeletalVsSingleBody (USkeletalMeshComponent* SkeletalMesh, const TArray<FName> SkeletalMeshBones, UPrimitiveComponent* SingleBody)	Re-enable collisions between a SkeletalMeshComponent and a PrimitiveComponent. You have to specify which bones in the SkeletalMeshComponent should be considered when enabling collisions.
void	EnableCollision_SkeletalVsSkeletal (USkeletalMeshComponent* SkeletalMeshA, const TArray<FName> SkeletalMeshBonesA, USkeletalMeshComponent* SkeletalMeshB, const TArray<FName> SkeletalMeshBonesB)	Re-enable collisions between two SkeletalMeshComponents. Specify which bones in both components should be considered when enabling collisions.
bool	IsCollisionDisabled (UPrimitiveComponent* ComponentA, UPrimitiveComponent* ComponentB, FName BoneNameA = NAME_None, FName BoneNameB = NAME_None)	Check that a given pair of bodies has a disabled collision.

void	GatherAllPhysicalComponents (AActor* Actor, TArray<UPrimitiveComponent*>& OutComponents)	Return a list of all components in given actor that have physics representation
void	GatherAllPhysicalBones (USkeletalMeshComponent* SkeletalMesh, TArray<FName>& OutBones)	Return a list of all bones that have any physics representation
void	GatherAllPhysicalBonesFrom (USkeletalMeshComponent* SkeletalMesh, FName StartBone, bool bIncludeStartBone, TArray<FName>& OutBones)	Returns a list of physical bones from a given bone

Performance

Collision Disabler is very efficient. The performance cost is Zero for objects that are not managed by CollisionDisabler. And a very small constant cost for every collision with objects that have a disabled collision. This means that only the number of actual colliding objects have an influence on performance.

Interaction Matrix

The table below shows whether the Collision Disabler can disable collisions.

Static Geometry - Not movable geometry e.g. buildings, rocks

Kinematic Body - Movable objects without enabled physics e.g. characters, projectiles

Dynamic Body - Simulate by physics object (enabled Simulate Physics) e.g. ragdolls

Body vs Body	Static Geometry	Kinematic Body	Dynamic Body
Static Geometry	N/A	No ¹	Yes
Kinematic Body	No ¹	No ¹	Yes ²
Dynamic Body	Yes	Yes ²	Yes

¹ - Kinematic and Static bodies don't generate collisions (Hit Events) because they are not simulated by the Physics Engine. For some object classes like characters or projectiles, you can use IgnoreActorWhenMoving or IgnoreComponentWhenMoving to prevent blocking objects.

² - Physical collision will be disabled, but still, kinematic objects may be blocked. In this case, also use IgnoreActorWhenMoving or IgnoreComponentWhenMoving nodes along with DisableCollision.